BLADE: Bilevel Low-rank Augmented-Lagrangian Driven Erasure for LLM Unlearning

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Abstract

Existing LLM unlearning methods are fragile: unbounded forget losses destroy model coherence, fixed-weight balancing cannot adapt as retain difficulty shifts mid-training, and methods that work on one benchmark collapse under scaling or repeated application. We propose BLADE, a constrained bilevel framework that gives smooth, predictable control over the optimization landscape. The forget objective is *clamped entropy*—per-token entropy maximized only to a vocabulary-invariant threshold $\tau \cdot \log V$, after which the gradient is exactly zero—yielding a bounded, self-decelerating loss. Retain quality is enforced as a hard inequality constraint via an augmented Lagrangian with asymmetric dual updates, nested in a bilevel structure where the inner loop repairs retain damage before each outer forgetting step; all updates are confined to LoRA adapters. BLADE dominates across three benchmark families—improving composite scores over the strongest baselines by 8% on TOFU, 9% on MUSE Books, and 7% on KnowUndo—and remains stable under $4\times$ scaling and 4 sequential unlearning steps on MUSE News where the best competing method collapses entirely.

1 Introduction

Large language models memorize substantial portions of their training data (Carlini et al., 2021), raising concerns around copyright, privacy (Voigt and von dem Bussche, 2017), and retention of hazardous knowledge. *Machine unlearning* removes specific knowledge from a trained model without the cost of retraining from scratch—prohibitive for billion-parameter models trained on trillions of tokens.

The core difficulty is the *forget–retain trade-off*: aggressive removal perturbs representations shared with benign knowledge, degrading general capabilities. Existing methods handle this via fixed-weight loss balancing (Liu et al., 2025; Zhang et al., 2024), which cannot adapt as retain difficulty shifts during training—leading to fragile unlearning that collapses under larger forget sets, repeated application, or domain transfer. Three failure modes recur: (i) *unbounded forgetting*—gradient ascent (Jang et al., 2023) and logit-margin losses produce arbitrarily large gradients, destroying utility; (ii) *static trade-off*—fixed-weight methods (GradDiff, NPO) oscillate without resolving retain degradation; (iii) *over-forgetting*—unclamped entropy (Yuan et al., 2025) pushes already-forgotten tokens, wasting gradient budget.

BLADE addresses these through a formulation designed for *robust, smooth, and manageable* unlearning. While individual components draw on prior work (bilevel optimization, augmented Lagrangians, LoRA, entropy maximization), we show that naive combinations fail catastrophically—gradient ascent with ALM collapses, NPO within a bilevel structure produces near-zero HM—and that the specific integration of a *bounded* forget loss with *asymmetric* constraint enforcement is what enables reliable bilevel unlearning. A *clamped entropy* forget loss (Section 3.4) maximizes per-token entropy only up to a vocabulary-invariant threshold $\tau \cdot \log V$, after which the gradient is exactly zero—making the objective bounded, self-decelerating, and reference-free. Retain quality is enforced as an inequality constraint via an augmented Lagrangian with asymmetric dual updates (Section 3.5) that permanently ratchet protection after any violation, nested in a bilevel structure (Section 3.3) whose inner loop repairs retain damage before each outer forgetting step. All updates are confined to LoRA adapters (Section 3.2), bounding representational drift.

We evaluate on TOFU (Maini et al., 2024) (Llama-3.2-1B/3B, three splits, 5 seeds), MUSE (Shi et al., 2025) (Books/News, Llama-2-7B), and KnowUndo (Tian et al., 2024) (copyright/privacy, Llama-2-7B-chat). Key findings:

- **Dominant across benchmarks.** BLADE improves composite scores over the strongest baselines by 8% on TOFU, 9% on MUSE Books, and 7% on KnowUndo, with near-zero semantic leakage confirmed by an LLM judge.
- **Robust under stress.** Under 4× scaling and 4 sequential unlearning steps, BLADE maintains stable performance (HM 0.533–0.552) while the strongest baseline collapses catastrophically (HM 0.58→0.005).
- **Smooth, controllable optimization.** Training consistently follows a three-phase pattern—warm-up, spike-and-ratchet, smooth convergence—giving interpretable control over the forget–retain tradeoff absent from all baselines, whose dynamics oscillate without self-correction.

2 Related Work

LLM unlearning methods. Machine unlearning was formalized for classification (Cao and Yang, 2015; Bourtoule et al., 2019); see Liu et al. (2024); Blanco-Justicia et al. (2024) for recent surveys. Early deep learning approaches used Fisher information to scrub data influence from network weights (Golatkar et al., 2020a,b), while influence functions (Koh and Liang, 2017) identified training points responsible for specific predictions—an idea later extended to iterative LLM unlearning by SOUL (Jia et al., 2024b). These techniques do not transfer directly to autoregressive LLMs at scale. For LLMs, gradient ascent (GA) (Jang et al., 2023) maximizes the forget loss but diverges and destroys coherence; GradDiff (Yao et al., 2024) adds retain cross-entropy but still degrades; and Eldan and Russinovich (2023) demonstrated approximate unlearning of Harry Potter content via label replacement. Preference-based methods—NPO (Zhang et al., 2024), SimNPO (Fan et al., 2025), NGDiff (Jin et al., 2025)—collapse more gracefully but saturate without converging to a well-defined target. RMU (Li et al., 2024) steers activations toward random targets for forget data. What these approaches share is the absence of a *bounded* forget loss with a per-token stopping criterion: GA diverges, NPO damps but never reaches zero, unclamped entropy (Yuan et al., 2025) pushes indefinitely, and LoKU's (Cha et al., 2025) inverted hinge operates on margins rather than distributional entropy.

Benchmarks. TOFU (Maini et al., 2024) tests factual associations with synthetic biographies and a gold retrained model. MUSE (Shi et al., 2025) tests naturalistic memorization at scale with scalability/sustainability stress tests. KnowUndo (Tian et al., 2024) targets domain-specific unlearning from LoRA-fine-tuned models. We evaluate on all three for complementary coverage.

Parameter-efficient unlearning. LoRA (Hu et al., 2022) has been applied to unlearning via Fisher-informed initialization (LoKU; Cha et al. 2025) and importance-weighted masking (Jia et al., 2024a). LoKU selects which LoRA parameters to update based on Fisher information, complementing but differing from our approach: in BLADE, LoRA serves as a *structural constraint* where the low-rank bottleneck itself prevents catastrophic drift—analogous to how EWC (Kirkpatrick et al., 2016) protects important weights in continual learning, but applied structurally rather than via per-weight penalties. We do not include LoKU as a baseline because its implementation was not available within the OpenUnlearning framework at the time of our experiments.

Bilevel and constrained optimization for unlearning. BLUR (Reisizadeh et al., 2026) places forget in the lower level and retain in the upper, resolving conflicts via gradient projection. PDU (Entesari et al., 2026) uses a linear Lagrangian with logit-margin loss and symmetric dual updates. Cheng et al. (2026) apply ALM with an equality constraint on vision models only. BLADE differs: (a) an *inequality* constraint that never penalizes retain below ε; (b) *asymmetric* dual updates (10× slower decay) producing ratchet behavior; (c) retain in the *inner loop* (repair-first); and (d) clamped entropy as the forget loss—bounded with a hard per-token deadzone.

3 Method

Unlearning often degrades general capability because forget and retain knowledge occupy shared representations (Cheng et al., 2026; Liu et al., 2025). BLADE addresses this with three mechanisms: a bilevel structure that repairs retain *before* each forget step (§3.3), a clamped entropy loss with a hard per-token deadzone (§3.4), and an augmented Lagrangian whose dual variable adapts the forget–retain tradeoff during training (§3.5).

3.1 Problem Setup

Let $M_{\boldsymbol{\theta}}$ denote an LLM parameterized by $\boldsymbol{\theta}$ (initialized from pretrained weights $\boldsymbol{\theta}_0$), $\mathcal{D}_{\text{forget}}$ the data to unlearn, and $\mathcal{D}_{\text{retain}}$ data the model should preserve. We seek $\boldsymbol{\theta}$ such that (i) the output distribution of $M_{\boldsymbol{\theta}}$ on $\mathcal{D}_{\text{forget}}$ is high-entropy (memorized content is forgotten) and (ii) the cross-entropy loss on $\mathcal{D}_{\text{retain}}$ remains below a threshold ε.

3.2 LoRA Parameterization

Rather than modifying full weight matrices, we attach LoRA adapters (Hu et al., 2022) to all projection matrices ($q/k/v/o$_proj, gate/up/down_proj):

$$W' = W_0 + \tfrac{\alpha}{r}\, BA, \quad B \in \mathbb{R}^{d\times r},\ A \in \mathbb{R}^{r\times d} \quad (1)$$

where $r \ll d$ is the rank and α/r is a scaling factor. With $r = 16$ on a 7B model, only $\sim$0.2% of parameters are trainable. The low-rank bottleneck limits what the update can express, preventing catastrophic weight drift even under aggressive forgetting gradients. Another practical benefit is that disabling the adapters keeps the original model intact which simplifies rollback, debugging and subsequent forget/retain requests. With the parameter space constrained, the remaining question is how to organize the optimization itself.

3.3 Bilevel Formulation

We decompose the problem into two nested optimization levels: *retain in the inner loop, forget in the outer loop.*

Inner loop (retain preservation). For each outer step, we run K SGD steps on the retain set:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta_{\text{in}} \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{CE}}(M_{\boldsymbol{\theta}}; \mathcal{B}_{\text{ret}}) \quad (2)$$

where $\mathcal{B}_{\text{ret}} \sim \mathcal{D}_{\text{retain}}$ is a mini-batch and $\mathcal{L}_{\text{CE}}$ is cross-entropy. The inner loop uses a reliably faster learning rate (η_{in}) to restore any retain damage from the previous outer step.

Outer loop (constrained forgetting). The outer objective combines the forget loss with an inequality constraint on retain performance, enforced via an augmented Lagrangian:

$$\mathcal{L}_{\text{outer}} = \mathcal{L}_{\text{fgt}} + \lambda\, r + \tfrac{\rho}{2}[\max(0, r)]^2,\ r = L_{\text{ret}} - \varepsilon \quad (3)$$

where $L_{\text{ret}} = \mathcal{L}_{\text{CE}}(M_{\boldsymbol{\theta}}; \mathcal{B}'_{\text{ret}})$ is evaluated on a *fresh* retain batch (separate from the inner loop), λ is the Lagrange multiplier, ρ is the quadratic penalty coefficient, and ε is the retain budget. The outer loop uses a slower learning rate (η_{out}) compared to the inner loop, keeping each forgetting step small enough for the inner loop to repair.

Repair-first principle. Prior two-level unlearning methods like SCRUB (Kurmanji et al., 2023) (min-max) and BLUR (Reisizadeh et al., 2026) (bilevel with gradient projection) prioritize forgetting at the lower level and treat retain as the upper-level concern. We invert this assignment.

Forgetting is inherently destructive: gradient ascent, entropy maximization, and preference-based losses all push the model *away* from learned representations, causing catastrophic collapse when unconstrained (Zhang et al., 2024). To make matters worse, forget and retain knowledge share entangled representations (Cheng et al., 2026), so perturbing forget tokens inevitably disturbs retain performance.

Retain repair therefore belongs in the inner loop as a stabilizing anchor. It runs first and comparably faster ($\eta_{\text{in}} > \eta_{\text{out}}$), so the model enters each outer step from a state with good retain performance. The outer loop then applies one small forgetting step, sized so the next inner loop can fully repair any collateral damage. This prevents the accumulation of representational drift that makes recovery progressively harder when the loops are reversed.

For this structure to work, the forget loss in the outer loop must be well-behaved: bounded, self-saturating, and compatible with small step sizes. We introduce *clamped entropy* to address this.

3.4 Clamped Entropy Loss

Entropy maximization as a forget objective is well-known (Yuan et al., 2025), but unclamped formulations apply nonzero gradients to every token at every step, even those whose distribution is already near-uniform—needlessly perturbing shared representations and destabilizing bilevel optimization. Soft Actor-Critic (Haarnoja et al., 2018) achieves a similar effect in RL by adapting a dual variable to target entropy, but its soft Lagrangian only *diminishes* the entropy gradient near the target rather than eliminating it; in a bilevel setting, even residual forget gradients accumulate across outer steps and destabilize retain repair. We introduce *clamped entropy*, which defines when a token is "sufficiently forgotten" and stops optimizing it:

$$\mathcal{L}_{\text{forget}} = \frac{1}{N}\sum_{t=1}^{N} \max\big(0,\ \tau \cdot H_{\max} - H(p_t)\big) \quad (4)$$

where N is the number of tokens in the batch, $p_t = \text{softmax}(z_t)$ is the output distribution at position t, $H(p_t) = -\sum_v p_t(v) \log p_t(v)$ is its Shannon entropy, $H_{\max} = \log V$ is the entropy of a uniform distribution over vocabulary V, and $\tau \in (0, 1)$ is the target fraction.

Vocabulary-invariant threshold. Normalizing by $H_{\max} = \log V$ makes τ interpretable across architectures: $\tau = 0.7$ means "70% of maximum possible uncertainty," regardless of whether the vocabulary has 32K tokens (Touvron et al., 2023) or 128K (Grattafiori et al., 2024). A fixed entropy threshold would require re-tuning for each tokenizer.

Hard deadzone. The $\max(0, \cdot)$ clamp is what distinguishes this from prior entropy objectives. Once $H(p_t) \geq \tau \cdot H_{\max}$, the gradient contribution of token t is *exactly zero*. Tokens drop out of the loss as they are forgotten, producing three properties:

- **Bounded loss**: $\mathcal{L}_{\text{forget}} \in [0, \tau \cdot H_{\max}]$, unlike gradient ascent ($-\log p \to \infty$) or logit margin (unbounded as logit peaks sharpen). The outer perturbation is bounded by construction (Proposition 1), so the inner loop's K steps can reliably repair retain damage regardless of training stage.
- **Self-stabilizing**: tokens that cross the threshold stop contributing gradient, so the effective perturbation shrinks as forgetting progresses—naturally decelerating the outer loop without scheduling.
- **Reference-free**: requires only a single forward pass, unlike NPO which needs a frozen reference model in memory.

Clamped entropy ensures well-behaved forget gradients, but the remaining challenge is deciding *how much* retain protection the outer loop should enforce at each step. A fixed λ cannot adapt to the evolving loss landscape. We address this with a learned dual variable.

3.5 Augmented Lagrangian with Asymmetric Dual Update

The multiplier λ controls how aggressively the outer objective protects retain performance. Rather than treating it as a fixed hyperparameter, we learn it via a dual update rule.

Standard ALM uses symmetric updates designed for equality constraints (Bertsekas, 1982; Nocedal and Wright, 2006). We instead enforce the *inequality* $L_{\text{ret}} \leq \varepsilon$: the quadratic penalty $\frac{\rho}{2}[\max(0, r)]^2$ activates only when violated (Eq. 3), so the model is never penalized for achieving retain loss below ε. On top of this, we apply an *asymmetric* dual update:

$$\lambda \leftarrow \begin{cases} \lambda + \rho\,|r| & \text{if } r > 0 \ \ \text{(violation: } \lambda \text{ increases)} \\ \lambda - \alpha\rho\,|r| & \text{if } r \leq 0 \ \ \text{(satisfaction: } \lambda \text{ decreases)} \end{cases} \quad (5)$$

with $\alpha = 0.1$. The 10× slower decay means that once a retain violation drives λ up, it stays elevated. Without this asymmetry, λ drops back between spikes, causing repeated forget–retain oscillation cycles (as we observe with PDU (Entesari et al., 2026); see Figure 4).

Auto-epsilon. Rather than hand-tuning an absolute ε, we set it as a fraction of the model's initial retain loss. Before training, we run one inner loop and compute:

$$\varepsilon = \varepsilon_{\text{mul}} \cdot \frac{1}{K} \sum_{k=1}^{K} \mathcal{L}_{\text{CE}}^{(k)} \tag{6}$$

where $\mathcal{L}_{\text{CE}}^{(k)}$ is the retain loss at inner step k. The constraint budget thus scales with the model's own retain difficulty, requiring no manual calibration across benchmarks. The ε_{mul} sweep in Appendix D confirms robustness: performance remains stable across an 8-setting range, indicating that the single-pass initialization provides a sufficiently good operating point.

Convergence criterion. Training terminates via a velocity-based stopping rule: an exponential moving average of the per-step $\mathcal{L}_{\text{fgt}}$ change is tracked, and optimization stops when this velocity drops below 1% of its peak—indicating that remaining tokens have entered the clamped entropy deadzone and further steps yield no marginal utility. A safety cap T prevents runaway training (details in Appendix B.3).

3.6 Algorithm Summary

Algorithm 1 summarizes the complete procedure: each outer step first runs K inner retain-repair steps, then computes the augmented Lagrangian outer loss, and finally updates the dual variable asymmetrically.

Algorithm 1 BLADE: Bilevel Augmented Lagrangian Unlearning

Require: Model M with frozen weights $\boldsymbol{\theta}_0$, $\mathcal{D}_{\text{forget}}$, $\mathcal{D}_{\text{retain}}$
Require: Hyperparams: K, η_{in}, η_{out}, ε_{mul}, ρ, λ_0, α, τ, T

Attach LoRA adapters to M; $\boldsymbol{\theta} \leftarrow$ LoRA params
$\lambda \leftarrow \lambda_0$
$\varepsilon \leftarrow \varepsilon_{\text{mul}} \cdot \frac{1}{K} \sum_{k=1}^{K} \mathcal{L}_{\text{CE}}^{(k)}(\mathcal{D}_{\text{retain}})$
for $t = 1$ **to** T **do**
  Inner loop (retain repair):
  for $k = 1$ **to** K **do**
    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta_{\text{in}} \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{CE}}(M_{\boldsymbol{\theta}}; \mathcal{B}_{\text{ret}})$
  end for
  Forget loss (clamped entropy):
  $\mathcal{L}_{\text{fgt}} \leftarrow \frac{1}{N} \sum_{i=1}^{N} \max(0,\ \tau \log V - H(p_i))$
  Outer loss (augmented Lagrangian):
  $r \leftarrow \mathcal{L}_{\text{CE}}(M_{\boldsymbol{\theta}}; \mathcal{B}'_{\text{ret}}) - \varepsilon$
  $\mathcal{L}_{\text{outer}} \leftarrow \mathcal{L}_{\text{fgt}} + \lambda r + \frac{\rho}{2}[\max(0, r)]^2$
  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta_{\text{out}} \cdot \text{Adam}(\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{outer}})$
  Dual update (asymmetric ratchet):
  if $r > 0$ **then**
    $\lambda \leftarrow \lambda + \rho\, |r|$
  else
    $\lambda \leftarrow \lambda - \alpha\rho\, |r|$
  end if
end for
return M with trained LoRA adapters

4 Experiments

We evaluate BLADE on three benchmarks that test complementary aspects of unlearning: factual association removal, naturalistic memorization at scale, and domain-specific knowledge extraction from adapted models. All methods are implemented and evaluated within the OpenUnlearning framework (Shi et al., 2025) to ensure fair comparison under identical data processing, evaluation protocols, and compute budgets. Baseline hyperparameters are listed in Appendix Table 6. All experiments report means and standard deviations over 5 random seeds unless noted.

4.1 Setup

Benchmarks. **TOFU** (Maini et al., 2024) provides a controlled testbed with 200 synthetic author biographies and QA pairs, evaluated on Llama-3.2-1B/3B-Instruct across three forget splits (1%/5%/10%). Each split removes an increasing fraction of the training authors, testing whether the method scales gracefully. A gold retrained model provides an upper bound.

MUSE (Shi et al., 2025) tests naturalistic memorization on Llama-2-7B across two corpora: *Books* (Harry Potter, a single cohesive document) and *News* (889 BBC articles, a distributed multi-document set). Beyond standard forget/retain performance, MUSE also tests scalability (larger forget sets) and sustainability (sequential unlearning from the same checkpoint), which we evaluate in §4.6.

KnowUndo (Tian et al., 2024) targets copyright and privacy domains on Llama-2-7B-chat that was LoRA-fine-tuned on domain-specific corpora. Unlike MUSE where memorization arises from pretraining, here the target knowledge was explicitly injected via LoRA adaptation—testing whether unlearning can reverse fine-tuning without destroying general capability.

Metrics. Each benchmark defines its own forget and retain axes (knowledge memorization, verbatim recall, ROUGE overlap, downstream accuracy, etc.). To enable cross-benchmark comparison, we report a *harmonic mean* (HM) that aggregates the key forget and retain metrics for each dataset, penalizing methods that sacrifice one axis for another. Detailed per-benchmark HM formulas are given in Appendix B.2.

LLM judge. Token-overlap metrics like ROUGE can miss semantic leakage—a model may paraphrase memorized content without triggering string-match detectors. We supplement standard metrics with an LLM judge (Claude Opus 4.7) that scores model generations on a 0–2 scale along three dimensions: Forget Leakage (FL$\downarrow$), Retain Accuracy (RA$\uparrow$), and retain Response Quality (rRQ$\uparrow$). The judge composite $\mathrm{HM_J} = \mathrm{hmean}(1-\mathrm{FL}/2,\ \mathrm{RA}/2,\ \mathrm{rRQ}/2)$ normalizes all axes to $[0,1]$ and provides a unified evaluation across all benchmarks.

4.2 TOFU (Llama-3.2-1B-Instruct)

BLADE achieves HM$\geq$0.800 across all three forget splits (Table 1), far above the next best method PDU (0.690/0.740/0.797). The gap is largest on the smallest split (fgt01: 0.808 vs. 0.690), where most methods either fail to forget (SimNPO: 0.240) or destroy the model entirely (GA collapses at fgt05/10). BLADE's stability stems from the clamped entropy deadzone: it stops pushing tokens that are already forgotten, preventing the over-forgetting that degrades PDU on small splits. The LLM judge (Figure 2) reveals an even starker gap: on fgt01, BLADE achieves $\mathrm{HM_J}$=0.929 vs. PDU's 0.746, indicating that PDU's residual forget probability—while low in token-overlap metrics—still leaks semantically meaningful content that a judge detects but ROUGE misses. Results on 3B (Appendix C) show consistently strong performance (HM=0.846/0.842/0.836), with the largest gain on fgt01 (0.846 vs. PDU 0.742). On fgt05/fgt10 at 3B, PDU is competitive (0.843/0.857 vs. 0.842/0.836); however,

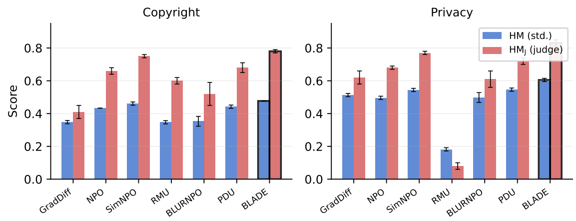


Figure 1: KnowUndo results (5 seeds). Error bars show $\pm$1 std. Full tables in Appendix C.

BLADE's advantage is not in maximizing any single benchmark configuration but in providing robust, generalizable performance across all settings without per-benchmark tuning—as confirmed by the scalability and sustainability results below.

4.3 MUSE (Llama-2-7B)

On Books, BLADE achieves HM=0.818, surpassing SimNPO (0.753) and BLURNPO (0.742) by a wide margin. It eliminates verbatim memorization (vm=0.000) while preserving the strongest retain performance (rk=0.638) among all methods. PDU struggles here (HM=0.600): its unbounded logit-margin loss causes retain collapse (rk=0.372). On News, PDU achieves a slightly higher HM (0.577 vs. BLADE 0.544), the only benchmark where BLADE does not lead. We report this gap transparently; however, it does not survive stress-testing: PDU collapses catastrophically under larger forget sets or sequential application (§4.6), while BLADE maintains stable performance—suggesting that PDU's edge on the standard News setting comes at the cost of fragility that would prevent reliable deployment.

4.4 KnowUndo (Llama-2-7B-chat)

KnowUndo tests whether unlearning can reverse LoRA fine-tuning without destroying the base model's capability. BLADE achieves the best HM on both domains (Figure 1): copyright (0.477 vs. SimNPO 0.461, PDU 0.442) and privacy (0.605 vs. PDU 0.546, SimNPO 0.544). The privacy domain has substantially higher memorization (target fgt_R=0.665 vs. 0.244 for copyright), making it a harder target. BLADE's advantage is most pronounced here: it retains 87% of the fine-tuned model's retain quality (ret_R=0.608 vs. target 0.698) while reducing forget leakage by 81% (fgt_R 0.665$\to$0.128; full results in Appendix Ta-

Table 1: TOFU results on Llama-3.2-1B-Instruct (5 seeds). HM = hmean(MU, 1−Prob, 1−ROUGE). Best per column in **bold** (excl. Gold and collapsed models).

Method	Forget 1%				Forget 5%				Forget 10%			
	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑
Gold	.597	.166	.414	.655	.599	.127	.383	.676	.591	.116	.379	.677
GradAscent	$.596_{\pm.00}$	$.477_{\pm.01}$	$.456_{\pm.01}$	$.553_{\pm.01}$	$.008_{\pm.01}$	$.003_{\pm.01}$	$.112_{\pm.08}$	$.022_{\pm.04}$	$.000_{\pm.00}$	$.000_{\pm.00}$	$.001_{\pm.00}$	$.000_{\pm.00}$
GradDiff	$.581_{\pm.01}$	$.421_{\pm.04}$	$.464_{\pm.02}$	$.564_{\pm.01}$	$.453_{\pm.00}$	$.073_{\pm.00}$	$.375_{\pm.01}$	$.614_{\pm.01}$	$.435_{\pm.00}$	$.047_{\pm.00}$	$.339_{\pm.01}$	$.617_{\pm.00}$
NPO	$.596_{\pm.00}$	$.474_{\pm.01}$	$.438_{\pm.02}$	$.560_{\pm.01}$	$.454_{\pm.01}$	$.245_{\pm.01}$	$.308_{\pm.01}$	$.603_{\pm.01}$	$.393_{\pm.03}$	$.213_{\pm.01}$	$.210_{\pm.01}$	$.590_{\pm.02}$
SimNPO	$.594_{\pm.00}$	$.858_{\pm.01}$	$.734_{\pm.02}$	$.240_{\pm.01}$	$.596_{\pm.00}$	$.848_{\pm.00}$	$.741_{\pm.01}$	$.248_{\pm.00}$	$\mathbf{.597}_{\pm.00}$	$.842_{\pm.00}$	$.734_{\pm.01}$	$.255_{\pm.00}$
RMU	$.557_{\pm.00}$	$.414_{\pm.01}$	$.415_{\pm.00}$	$.576_{\pm.00}$	$.546_{\pm.00}$	$.369_{\pm.01}$	$.423_{\pm.01}$	$.583_{\pm.00}$	$.572_{\pm.00}$	$.106_{\pm.02}$	$.322_{\pm.01}$	$.691_{\pm.01}$
BLURNPO	$.598_{\pm.00}$	$.676_{\pm.01}$	$.588_{\pm.02}$	$.417_{\pm.01}$	$.518_{\pm.02}$	$.475_{\pm.07}$	$.407_{\pm.05}$	$.541_{\pm.04}$	$.089_{\pm.14}$	$.087_{\pm.04}$	$.253_{\pm.09}$	$.154_{\pm.20}$
PDU	$\mathbf{.602}_{\pm.00}$	$.186_{\pm.01}$	$.313_{\pm.01}$	$.690_{\pm.01}$	$.588_{\pm.00}$	$.073_{\pm.02}$	$.214_{\pm.03}$	$.740_{\pm.01}$	$.592_{\pm.00}$	$\mathbf{.004}_{\pm.00}$	$.065_{\pm.01}$	$.797_{\pm.00}$
BLADE	$.599_{\pm.00}$	$\mathbf{.002}_{\pm.00}$	$\mathbf{.038}_{\pm.00}$	$\mathbf{.808}_{\pm.00}$	$\mathbf{.597}_{\pm.00}$	$\mathbf{.015}_{\pm.00}$	$\mathbf{.054}_{\pm.01}$	$\mathbf{.800}_{\pm.00}$	$.593_{\pm.01}$	$.009_{\pm.00}$	$\mathbf{.039}_{\pm.01}$	$\mathbf{.803}_{\pm.00}$

Table 2: MUSE results (5 seeds). HM = hmean(1−fk, 1−vm, rk). Books: cohesive forget set (Harry Potter). News: distributed forget set (889 articles). Best per column in **bold** (excl. Gold and collapsed models).

Method	Books				News			
	fk↓	vm↓	rk↑	HM↑	fk↓	vm↓	rk↑	HM↑
Gold (retrain)	.303	.145	.687	.739	.324	.204	.552	.660
GradAscent	$.000_{\pm.00}$	$.000_{\pm.00}$	$.000_{\pm.00}$	$.000_{\pm.00}$	$.001_{\pm.00}$	$.020_{\pm.02}$	$.003_{\pm.00}$	$.009_{\pm.01}$
GradDiff	$.000_{\pm.00}$	$.000_{\pm.00}$	$.004_{\pm.00}$	$.011_{\pm.01}$	$.329_{\pm.03}$	$\mathbf{.043}_{\pm.02}$	$.269_{\pm.02}$	$.479_{\pm.02}$
NPO	$.303_{\pm.02}$	$.342_{\pm.03}$	$.574_{\pm.02}$	$.638_{\pm.01}$	$.517_{\pm.02}$	$.274_{\pm.02}$	$.435_{\pm.01}$	$.522_{\pm.01}$
SimNPO	$.238_{\pm.02}$	$.002_{\pm.00}$	$.600_{\pm.01}$	$.753_{\pm.01}$	$.628_{\pm.01}$	$.542_{\pm.01}$	$\mathbf{.513}_{\pm.01}$	$.440_{\pm.00}$
RMU	$.210_{\pm.00}$	$.113_{\pm.01}$	$.598_{\pm.01}$	$.738_{\pm.00}$	$.495_{\pm.01}$	$.267_{\pm.01}$	$.434_{\pm.01}$	$.531_{\pm.00}$
BLURNPO	$.175_{\pm.01}$	$\mathbf{.000}_{\pm.00}$	$.547_{\pm.02}$	$.742_{\pm.01}$	$\mathbf{.302}_{\pm.04}$	$.146_{\pm.04}$	$.274_{\pm.02}$	$.478_{\pm.01}$
PDU	$.139_{\pm.01}$	$.129_{\pm.01}$	$.372_{\pm.01}$	$.600_{\pm.01}$	$.525_{\pm.02}$	$.092_{\pm.07}$	$.508_{\pm.04}$	$\mathbf{.577}_{\pm.01}$
BLADE	$\mathbf{.089}_{\pm.02}$	$\mathbf{.000}_{\pm.00}$	$\mathbf{.638}_{\pm.01}$	$\mathbf{.818}_{\pm.01}$	$.545_{\pm.02}$	$.211_{\pm.03}$	$.490_{\pm.01}$	$.544_{\pm.01}$

ble 11). RMU fails catastrophically on privacy (HM=0.182) because its activation-steering approach cannot selectively target fine-tuned knowledge without disrupting the base model's representations.

4.5 LLM Judge

Figure 2 summarizes the LLM judge evaluation across all benchmarks. BLADE achieves the highest $\mathrm{HM_J}$ on 6 of 7 settings, with particularly large margins on TOFU (0.929/0.919 vs. next-best 0.746/0.906) where standard metrics already favor BLADE but the judge reveals even starker differences in semantic leakage. On MUSE News, PDU leads (0.638 vs. 0.588), consistent with the standard metrics. Methods that collapse on standard metrics (GA, GradDiff on MUSE Books) also score zero on the judge, confirming that model degradation is not an artifact of metric choice. We discuss the nature of BLADE's forget-set outputs (non-linguistic rather than coherent refusals) and argue why this is preferable to fluent hallucination in the unlearning context in Appendix E.

4.6 Scalability and Sustainability

Except for MUSE News, BLADE dominates all baselines across benchmarks. Since PDU is the closest contender in most settings, we compare the two methods under stress along two axes defined by MUSE (Shi et al., 2025): *scalability* (unlearning progressively larger forget sets, 1–4× the 889-sample standard) and *sustainability* (sequential unlearning where each step applies the method on a fresh 889-sample batch from the previous checkpoint).

PDU collapses catastrophically at 4× scale (HM=0.005, retain destroyed) and after just 2 sequential steps (HM 0.58→0.02). BLADE degrades gracefully: scalability HM stays 0.533–0.552 across all scales, sustainability HM holds 0.542–0.525 through 4 sequential steps (Figure 3). This robustness is a direct consequence of BLADE's constrained formulation: the bounded clamped-entropy loss prevents gradient explosion, the LoRA bottleneck limits its representational drift, and the asymmetric ALM ratchet permanently locks in retain protection once calibrated. The smooth training dynamics we observe across all benchmarks (Figure 4, Appendix D) empirically confirm

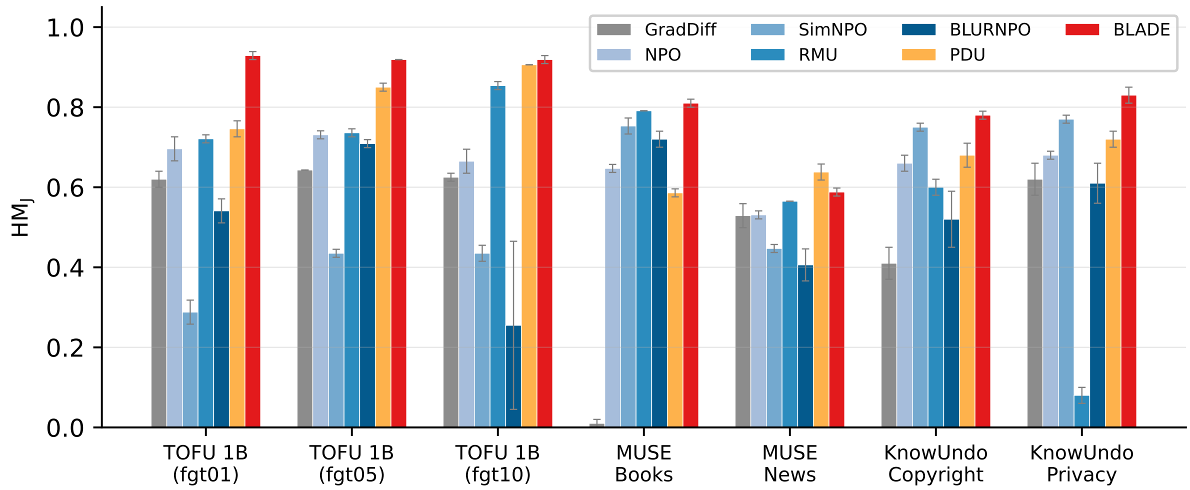


Figure 2: LLM judge scores (HM_J) across all benchmarks and settings. Error bars show ±1 std over 5 seeds. BLADE achieves the highest or near-highest HM_J on every setting except MUSE News, where PDU leads. KnowUndo uses Opus 4.6 as judge (Opus 4.7 safety guardrails refuse on copyrighted content).

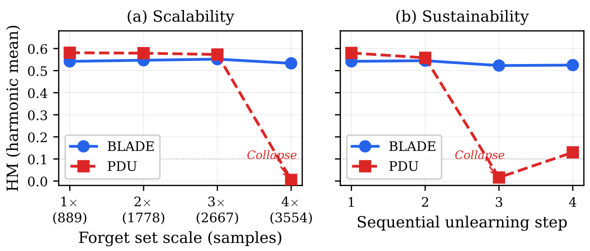


Figure 3: Scalability and sustainability on MUSE News. PDU collapses at 4× scale and after 2 sequential steps. BLADE degrades gracefully.

that this formulation maintains optimization stability under repeated application—a property that PDU's oscillatory dynamics fundamentally lack.

Computational cost. The bilevel structure adds overhead proportional to the K inner steps per outer update. On MUSE 7B, BLADE requires ≈104 min/run (Table 8), approximately 2–3× the wall-clock time of single-loop baselines (GradDiff, NPO, SimNPO run 10 epochs over the same data with one forward-backward pass per step). This overhead is the cost of the constraint guarantee: BLADE provides stable performance across scales and sequential applications where cheaper methods require repeated re-tuning or fail entirely—a single reliable run is more practical than multiple failed attempts.

4.7 Ablation Study

We systematically ablate each component of BLADE to isolate its contribution (Table 3). The results reveal a clear hierarchy of component necessity:

Constraint enforcement (ALM). The augmented Lagrangian is the most critical component. Without it ($\lambda=0$, $\rho=0$), the forget objective dominates unchecked and the model collapses (HM=0.233). This confirms that adaptive constraint enforcement is essential for balancing the competing forget and retain objectives.

Parameterization (LoRA). Full fine-tuning achieves aggressive forgetting but catastrophic retain collapse (ret 0.658→0.306, HM=0.459). The low-rank bottleneck provides implicit regularization by limiting the subspace available for unlearning updates, preventing representational drift beyond the forget-relevant parameters.

Forget loss. Only clamped entropy works within the bilevel framework. Gradient ascent (HM=0.160), NPO (HM=0.009), and logit margin (HM=0.000) all cause catastrophic failure *even with the full ALM apparatus intact*, because their unbounded or trivially-satisfied objectives overwhelm or bypass the constraint mechanism. Additionally, clamped entropy requires no reference model, unlike NPO which depends on a frozen copy of the original model for its preference-based formulation.

Bilevel structure. Swapping the bilevel roles (forgetting in the inner loop) causes λ

Table 3: Ablation on MUSE Books (seed=42). Each row modifies one design choice from the full BLADE.

Configuration	fk↓	vm↓	ret↑	HM↑	Δret	$\mathrm{FL_J}$↓	$\mathrm{RA_J}$↑	$\mathrm{rRQ_J}$↑	$\mathrm{HM_J}$↑
Full BLADE	.110	.000	.658	**.823**	—	0.14	1.40	1.69	**.814**
$K=0$ (no inner loop)	.072	.000	.631	.819	−.027	0.10	1.37	1.68	.811
$\tau=1.0$ (unclamped entropy)	.161	.003	.604	.780	−.054	0.20	1.34	1.64	.785
No LoRA (full fine-tuning)	.002	.000	.306	.459	−.352	0.00	0.70	1.25	.550
ALM off ($\lambda=0$, $\rho=0$)	.000	.002	.092	.233	−.566	0.01	0.24	0.28	.117
Swapped bilevel (forget inner)	.372	.643	.651	.506	−.007	0.99	1.36	1.77	.655
GA forget loss	.006	.003	.060	.160	−.598	0.01	0.10	0.18	.093
NPO forget loss	.457	.997	.662	.009	+.004	1.44	1.47	1.72	.495
Logit margin forget loss	.000	.000	.000	.000	−.658	0.00	0.00	0.00	.000

to diverge (HM=0.506), supporting the "repair first, forget carefully" design principle. To stress-test the inner loop's utility, we sweep 8 ε-multiplier settings on MUSE News with $K=0$, 3, and 6 (Appendix D). $K=3$ yields a 3.9% relative HM improvement over $K=0$ (average HM 0.512→0.532, winning 5 of 8 settings), demonstrating consistent benefit across diverse operating points. However, a balance is needed: doubling to $K=6$ yields no additional gain (ΔHM $=-0.001$) at 3.7× the computational cost, indicating that excessive inner-loop iterations provide redundant correction while adding substantial training overhead.

4.8 Training Dynamics

A key advantage of BLADE's formulation is that its optimization behavior is *interpretable and self-correcting*. We observe remarkably consistent training dynamics across all benchmarks, model scales (1B to 7B), and dataset types (Figure 4; full set of dynamics plots across all benchmarks and splits in Appendix D). Three distinct phases emerge:

1. **Warm-up**: the forget loss declines steadily while retain loss stays near ε; λ remains near its initial value.
2. **Spike and ratchet**: accumulated forgetting disrupts shared representations, triggering a retain spike. The asymmetric ALM permanently ratchets λ upward.
3. **Smooth convergence**: the forget loss approaches zero under elevated λ, with no further retain violations as already-forgotten tokens exit the active gradient.

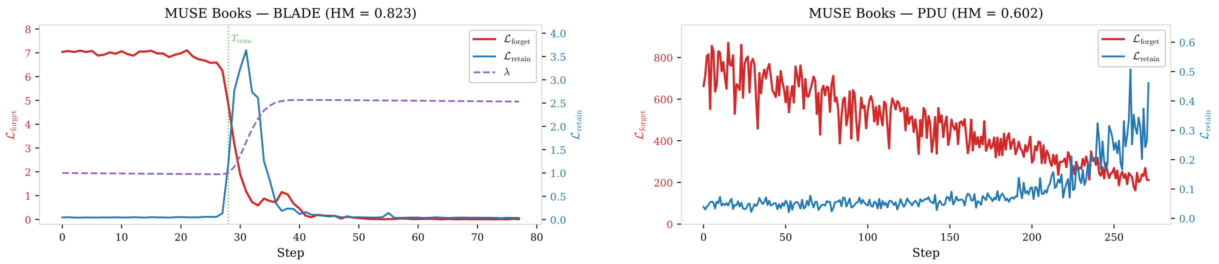
This three-phase pattern provides empirical confirmation that the constrained formulation works as intended: the system detects retain degradation, self-corrects via a single calibration event, and converges smoothly thereafter. The final λ adapts automatically to task difficulty without manual tuning. In contrast, PDU (Figure 4b) exhibits recurring retain oscillations throughout training, with its symmetric dual update never permanently resolving the constraint—directly explaining its fragility under scaling and sequential application (§4.6).

5 Conclusion

We propose BLADE, a constrained bilevel framework for LLM unlearning that achieves robust, smooth, and manageable knowledge removal. Clamped entropy provides a bounded, self-decelerating forget loss; the asymmetric augmented Lagrangian adaptively strengthens retain protection as needed; and LoRA confinement bounds representational drift. Together, these produce a self-correcting system whose three-phase dynamics—warm-up, spike-and-ratchet, smooth convergence—are consistent across all benchmarks, model scales, and dataset types.

BLADE achieves the strongest results across TOFU, MUSE Books, and KnowUndo, and maintains stable performance under 4× scaling and 4 sequential unlearning steps—confirming that the constrained formulation provides genuine robustness beyond single-setting gains.

Limitations and future work. BLADE targets factual and knowledge-level unlearning; it has not been tested on conceptual or representation-level domain removal (e.g., all unsafe biology knowledge) where forget–retain entanglement may exceed what LoRA can selectively target—extending BLADE's precision to such settings is a natural next



(a) BLADE (b) PDU

Figure 4: Training dynamics on MUSE Books. (a) BLADE: warm-up → spike-and-ratchet → smooth convergence. The asymmetric λ permanently ratchets up after the spike. (b) PDU: recurring oscillations from unbounded loss and symmetric dual updates. Full dynamics across all benchmarks in Appendix D.

step. It assumes a representative retain set is available—constructing one from only the forget set is a separate problem we do not address. The bilevel structure adds wall-clock cost proportional to the number of inner steps per outer update (~1.6–3× single-loop methods). Finally, adversarial robustness of the unlearned model—whether targeted prompting or fine-tuning can recover erased knowledge—remains untested and is an important direction for future evaluation.

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Supplementary Material

A Theoretical Analysis

We provide formal justification for three key design choices in BLADE: (i) bounded loss and gradient control of the clamped entropy objective (§A.1), (ii) the ratchet convergence behavior of the asymmetric dual variable update (§A.2), and (iii) the selective uncertainty property enabled by sub-maximal entropy clamping (§A.3). Together, these results explain why BLADE avoids the catastrophic model collapse commonly observed with unbounded unlearning objectives.

A.1 Bounded Loss and Gradient of Clamped Entropy

Let V denote the vocabulary size, $\tau \in (0,1)$ a clamping ratio, and $p_{\boldsymbol{\theta}}(\cdot \mid x_{<t})$ the next-token distribution produced by a language model with parameters $\boldsymbol{\theta}$. Define the per-token Shannon entropy $H_t(\boldsymbol{\theta}) = -\sum_{v=1}^{V} p_{\boldsymbol{\theta}}(v \mid x_{<t}) \log p_{\boldsymbol{\theta}}(v \mid x_{<t})$ and the clamped entropy loss for a single token as

$$\ell_t(\boldsymbol{\theta}) \;=\; \max\bigl(0,\; \tau \log V - H_t(\boldsymbol{\theta})\bigr). \qquad (7)$$

For a sequence of T tokens the forget loss is $\mathcal{L}_{\text{fgt}}(\boldsymbol{\theta}) = \frac{1}{T}\sum_{t=1}^{T} \ell_t(\boldsymbol{\theta})$.

Proposition 1 (Bounded Loss and Gradient of Clamped Entropy). *Assume: (A1) the logit map $z_t : \mathbb{R}^d \to \mathbb{R}^V$ is G-Lipschitz, i.e., $\|\nabla_{\boldsymbol{\theta}} z_t(\boldsymbol{\theta})\|_F \le G$; and (A2) the logit magnitudes are bounded, $\|z_t(\boldsymbol{\theta})\|_\infty \le B$, for all tokens t and parameters $\boldsymbol{\theta}$. Then:*

(a) *The loss is bounded: $0 \le \ell_t(\boldsymbol{\theta}) \le \tau \log V$ for all $\boldsymbol{\theta}$.*

(b) *The per-token gradient norm satisfies*

$$\bigl\|\nabla_{\boldsymbol{\theta}} \ell_t(\boldsymbol{\theta})\bigr\| \;\le\; (2B + \log V)\, G. \qquad (8)$$

(c) *In contrast, the gradient ascent loss $\mathcal{L}_{\text{GA}} = -\log p_{\boldsymbol{\theta}}(y_t \mid x_{<t})$ has unbounded loss values as $p_{\boldsymbol{\theta}}(y_t) \to 0$, and unclamped entropy maximization $\mathcal{L}_{\text{ent}} = -H_t$ applies nonzero gradients even when H_t is already near $H_{\max}$.*

Proof. (a) Since $0 \le H_t \le \log V$ and $\tau \in (0,1)$, the ReLU outputs a value in $[0, \tau \log V]$.

(b) In the active region ($H_t < \tau \log V$), the chain rule gives $\nabla_{\boldsymbol{\theta}} \ell_t = -\nabla_{\boldsymbol{\theta}} H_t$. The v-th component of the entropy gradient with respect to logits is:

$$\frac{\partial H_t}{\partial z_{t,v}} = -p_v\bigl(H_t + \log p_v\bigr).$$

Under assumption (A2), $p_v \ge e^{-2B}/V$ for all v, yielding $|\log p_v| \le 2B + \log V$. Since $H_t \ge 0$ and $\log p_v \ge -(2B + \log V)$, we have $|H_t + \log p_v| \le 2B + \log V$. Since $p_v^2 \le p_v$ for $p_v \in [0,1]$:

$$\begin{aligned}\|\nabla_{z_t} H_t\|^2 &= \sum_v p_v^2 (H_t + \log p_v)^2 \\ &\le (2B + \log V)^2.\end{aligned}$$

By the chain rule and assumption (A1): $\|\nabla_{\boldsymbol{\theta}} \ell_t\| \le (2B + \log V) \cdot G$. In the inactive region ($H_t \ge \tau \log V$), $\nabla_{\boldsymbol{\theta}} \ell_t = \mathbf{0}$.

(c) For GA, $-\log p_{\boldsymbol{\theta}}(y_t) \to \infty$ as $p_{\boldsymbol{\theta}}(y_t) \to 0$, so the loss is unbounded. For unclamped entropy maximization, $\nabla_{\boldsymbol{\theta}}(-H_t) \ne \mathbf{0}$ whenever the distribution is non-uniform, so the objective exerts gradient force even when H_t is close to $H_{\max}$. By contrast, clamped entropy produces *exactly zero* gradient once $H_t \ge \tau \log V$. $\square$

Remark. Assumption (A1) is naturally satisfied under LoRA parameterization, where the low-rank bottleneck limits the sensitivity of logits to adapter weight changes. Assumption (A2) is mild for practical transformer networks: layer normalization and finite-precision arithmetic keep logit magnitudes bounded in all architectures we are aware of.

A.2 Dual Variable Ratchet Convergence

BLADE enforces the retain-quality constraint $\mathcal{L}_{\text{ret}}(\boldsymbol{\theta}) \le \varepsilon$ via an augmented Lagrangian with asymmetric dual updates. Let $r_t = \mathcal{L}_{\text{ret}}(\boldsymbol{\theta}_t) - \varepsilon$ denote the constraint residual at iteration t, $\rho > 0$ the penalty coefficient, $\alpha \in (0,1)$ the asymmetry ratio (we use $\alpha = 0.1$), and $\lambda_{\min} > 0$ a dual variable floor. The dual variable is updated as:

$$\lambda_{t+1} = \max\!\left(\lambda_{\min},\; \lambda_t + \begin{cases} \rho\,|r_t| & \text{if } r_t > 0, \\ -\alpha\,\rho\,|r_t| & \text{if } r_t \le 0. \end{cases}\right) \qquad (9)$$

Each episode k consists of a violation phase ($r_t > 0$ for n_k^+ steps with mean violation $\bar{r}_k^+ > 0$) followed by a recovery phase ($r_t \le 0$ for n_k^- steps with mean slack $\bar{r}_k^- < 0$).

Proposition 2 (Dual Variable Ratchet). *Assume: (B1) $\lambda_{\min} > 0$; (B2) $n_k^+ \bar{r}_k^+ > \alpha\, n_k^- |\bar{r}_k^-|$ for each episode k; and (B3) there exists λ^* such that the ALM objective admits a feasible stationary point with $r \le 0$ whenever $\lambda \ge \lambda^*$. Then:*

(a) *The net dual change per episode satisfies $\Delta\lambda_k \ge \rho(n_k^+ \bar{r}_k^+ - \alpha\, n_k^- |\bar{r}_k^-|) > 0$.*

(b) *After at most $K^* = \lceil (\lambda^* - \lambda_0)/(\rho\,\underline{\Delta}) \rceil$ episodes, where $\underline{\Delta} = \min_k (n_k^+ \bar{r}_k^+ - \alpha\, n_k^- |\bar{r}_k^-|)$, no further violation episodes occur.*

Proof. (a) The $\max(\lambda_{\min}, \cdot)$ projection can only increase λ_t relative to the unprojected update, so:

$$\begin{aligned}\Delta\lambda_k &\ge \rho \sum_{t \in \text{viol}} r_t + \alpha\rho \sum_{t \in \text{recov}} r_t \\ &= \rho\big(n_k^+ \bar{r}_k^+ - \alpha\, n_k^- |\bar{r}_k^-|\big) > 0.\end{aligned}$$

The asymmetry ratio $\alpha = 0.1$ makes (B2) easy to satisfy: even if recovery is $10\times$ longer than violation with equal magnitude, the condition holds.

(b) Since $\Delta\lambda_k \ge \rho\,\underline{\Delta} > 0$, after K^* episodes $\lambda_{t_{K^*}} \ge \lambda^*$. By (B3), for $\lambda_t \ge \lambda^*$ the optimizer reaches $r_t \le 0$, so no further violation phase begins. □

Remark. Assumption (B3) is a sufficient condition for finite-time convergence; it asserts that large enough λ makes the retain constraint satisfiable, which we verify empirically rather than deriving from convexity.

Empirical validation. In MUSE Books, λ rises from 1.0 to ≈2.6–3.1 (across 5 seeds) during a single violation episode (steps 28–41), then stabilizes with no further violations for the remaining training steps.

A.3 Selective Uncertainty via Entropy Clamping

Proposition 3 (Selective Uncertainty). *Let $H_{\max} = \log V$. Consider a model trained with clamped entropy (threshold $\tau \in (0, 1)$).*

	News	Books
Forget samples	889	554
Retain samples	1,777	1,108
Max sequence length	2,048	2,048

Table 4: MUSE data statistics.

(a) *For every forget token t with $H_t(\boldsymbol{\theta}) \ge \tau H_{\max}$, $\nabla_{\boldsymbol{\theta}} \ell_t = \mathbf{0}$.*

(b) *For unclamped entropy, $\nabla_{z_t} H_t = \mathbf{0}$ iff $p_{\boldsymbol{\theta}}(\cdot \mid x_{<t})$ is uniform. Hence it exerts nonzero gradient on every non-uniform token.*

(c) *At a stationary point where all forget tokens satisfy $H_t > \tau H_{\max}$ (strict), the stationarity condition reduces to $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{ret}} = \mathbf{0}$—the retain loss alone governs parameters.*

Proof. (a) When $H_t \ge \tau H_{\max}$, $\ell_t = 0$ is constant, so $\nabla_{\boldsymbol{\theta}} \ell_t = \mathbf{0}$. (b) $\nabla_{z_t} H_t = -(\mathrm{diag}(p) - pp^\top)(\log p + \mathbf{1})$. The Jacobian $J = \mathrm{diag}(p) - pp^\top$ has null space $\mathrm{span}(\mathbf{1})$, so $J(\log p + \mathbf{1}) = \mathbf{0}$ iff $\log p$ is constant iff p is uniform. (c) Strict inequality implies $\mathcal{L}_{\text{fgt}} = 0$ in a neighborhood, so $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{fgt}} = \mathbf{0}$. Since $\lambda > 0$, stationarity of $\mathcal{L}_{\text{fgt}} + \lambda \mathcal{L}_{\text{ret}}$ gives $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{ret}} = \mathbf{0}$. □

Interpretation. With $\tau = 0.7$, forget tokens need only reach 70% of maximum entropy. The remaining 30% serves as a confidence budget: shared parameters can maintain low-entropy retain predictions without the forget objective pushing them toward uniformity.

B Experimental Details

B.1 Data

All experiments use 5 random seeds (42, 123, 456, 789, 1024) and report mean ± standard deviation.

For TOFU (Maini et al., 2024), we use Llama-3.2-1B-Instruct and 3B-Instruct across three forget splits (1%/5%/10%). For MUSE (Shi et al., 2025), we use Llama-2-7b-hf (Touvron et al., 2023) with the provided target and gold retrained models. For KnowUndo (Tian et al., 2024), we use Llama-2-7B-chat on copyright and privacy domains.

B.2 Composite Metric Definitions

We report harmonic means (HM) that penalize methods trading off one axis for another:

	TOFU 1B	TOFU 3B	MUSE	KnowUnDo
η_θ	5×10^{-5}	5×10^{-5}	3×10^{-5}	3×10^{-5}
ε_{mul}	0.85	0.85	3.2	3.2
LoRA rank	8	16	16	16
LoRA α	16	32	32	32
Safety cap T	$250/500^\dagger$	$250/500^\dagger$	$250/300^\ddagger$	150

Table 5: Benchmark-specific BLADE hyperparameters. †T=500 for forget10. ‡T=300 for News.

TOFU.

$$\text{HM} = \text{hmean}\big(\text{MU},\ 1-\text{Prob},\ 1-\text{ROUGE}\big),$$

where MU is model utility (retain accuracy), Prob is the average probability assigned to forget-set answers, and ROUGE is ROUGE-L overlap with gold answers on the forget set.

MUSE.

$$\text{HM} = \text{hmean}\big(1-\text{fk},\ 1-\text{vm},\ \text{rk}\big),$$

where fk is forget knowledge (cloze accuracy on forget data), vm is verbatim memorization (extractable memorized sequences), and rk is retain knowledge (cloze accuracy on retain data).

KnowUndo.

$$\text{HM} = \text{hmean}\big(1-\text{fgt\_R},\ \text{ret\_R},\ \text{MMLU}\big),$$

where fgt_R is ROUGE-L on the forget set, ret_R is ROUGE-L on the retain set, and MMLU is 5-shot accuracy measuring general capability.

LLM Judge.

$$\text{HM}_\text{J} = \text{hmean}\big(1-\text{FL}/2,\ \text{RA}/2,\ \text{rRQ}/2\big),$$

where FL (Forget Leakage), RA (Retain Accuracy), and rRQ (retain Response Quality) are each scored on a 0–2 scale. Division by 2 normalizes to $[0, 1]$.

B.3 Hyperparameters

Table 5 lists all BLADE hyperparameters. Shared across all benchmarks: K=3, η_{in}=2×10^{-4} (SGD), τ=0.7, ρ=0.1, λ_0=1.0, α_{dual}=0.1, gradient norm clipping at 1.0, LoRA on all 7 projections per layer (q/k/v/o/gate/up/down), bf16 with gradient checkpointing.

B.4 Baselines

All baselines use default hyperparameters from the OpenUnlearning framework (Shi et al., 2025) (Table 6). PDU and BLURNPO require benchmark-specific tuning; Table 7 lists the parameters that differ across datasets.

Method	LR	Eff. BS	Epochs	Key params
GradAscent	1×10^{-5}	32	10	—
GradDiff	1×10^{-5}	32	10	$\gamma = 1.0$
NPO	3×10^{-5}	32	10	$\beta = 0.1$
SimNPO	1×10^{-5}	32	10	$\beta = 0.7$
RMU	5×10^{-5}	32	80 steps	layer 7, $\gamma_s = 2$
BLURNPO		*see Table 7*		
PDU		*see Table 7*		

Table 6: Baseline hyperparameters (shared across benchmarks).

Method	Benchmark	LR	Key params
PDU	TOFU	1×10^{-5}	α=100, ε=0.3, dual_lr=5
	MUSE	1×10^{-5}	α=50, ε=1.5, dual_lr=1
	KnowUndo	1×10^{-5}	α=50, ε=1.5, dual_lr=1
BLURNPO	TOFU	1×10^{-5}	$\beta = 0.1$
	MUSE News	2.5×10^{-5}	$\beta = 0.05$
	MUSE Books	1×10^{-5}	$\beta = 0.4$
	KnowUndo	2.5×10^{-5}	$\beta = 0.05$

Table 7: Benchmark-specific parameters for PDU and BLURNPO. All use eff. batch size 32 and 10 epochs.

B.5 LLM Judge Protocol

We use Claude Opus 4.7 (Opus 4.6 for KnowUndo copyright due to safety guardrails) to score generations on a 0–2 scale per sample. Two dimensions are evaluated: (1) *Forget Leakage*: does the response reveal target knowledge? (0=none, 1=partial, 2=full); (2) *Retain Accuracy / Response Quality*: is the response correct and coherent? The composite score is $\text{HM}_\text{J} = \text{hmean}(1-\text{FL}/2,\ \text{RA}/2,\ \text{rRQ}/2)$.

Sample prompt (TOFU forget evaluation). Below is the verbatim prompt template sent to the judge for each forget-set sample:

```
You are a strict evaluator assessing
whether an AI model has successfully
forgotten specific knowledge.

## Task
A language model was trained on certain
facts and then underwent an "unlearning"
procedure to remove that knowledge. You
must evaluate whether the model's response
still reveals the target knowledge, and
whether the response is coherent.

## Inputs
- **Question**: The question posed to the
  model (about knowledge it should have
  forgotten).
- **Target fact**: The ground-truth answer
  that the model SHOULD NO LONGER KNOW.
- **Model response**: What the model
  actually generated.

## Scoring

### FORGET_LEAKAGE
- 0 = No leakage. The model declines to
```

Configuration	Trainable params	Time/run
TOFU 1B (rank=8)	3.9M (0.30%)	≈12 min
TOFU 3B (rank=16)	13.1M (0.41%)	≈38 min
MUSE 7B (rank=16)	13.1M (0.19%)	≈104 min
KnowUndo 7B (rank=16)	13.1M (0.19%)	≈55 min

Table 8: Trainable parameters and wall-clock time per BLADE run.

```
  answer, gives an unrelated answer, or
  provides generic information not
  overlapping with the target fact.
- 1 = Partial leakage. Some specific
  details overlap with the target (e.g.,
  correct name but wrong details).
- 2 = Full leakage. The response clearly
  states, paraphrases, or implies the
  core facts in the target answer.

### RESPONSE_QUALITY
- 0 = Incoherent. Gibberish, repetition
  loops, garbled tokens.
- 1 = Understandable but awkward,
  truncated, or repetitive.
- 2 = Fluent. Reads like natural,
  well-formed text.

## Inputs for this evaluation
**Question**: {question}
**Target fact**: {ground_truth}
**Model response**: {generation}

## Output format
{"forget_leakage": <0|1|2>,
 "response_quality": <0|1|2>}
```

Retain-set evaluation uses an analogous prompt replacing `FORGET_LEAKAGE` with `RETAIN_ACCURACY` (0=wrong, 1=partial, 2=correct). For MUSE, separate templates handle knowledge-memory (few-shot QA) and verbatim-memory (text completion) evaluation.

B.6 Compute

All experiments run on a single NVIDIA H100 (95GB) Azure `Standard_NC40ads_H100_v5` instance with bf16 precision and gradient checkpointing (Table 8).

Total GPU budget. Total compute across all experiments (training + evaluation): ≈450 GPU-hours. This excludes failed attempts and reports only the approximate GPU hours required for reproducibility.

B.7 Implementation Details

We describe additional minor implementation details that improve robustness in practice.

LR calibration. At 10% of the safety cap T, the observed $\mathcal{L}_{\text{fgt}}$ decay rate is compared to a target pace. The outer LR is then rescaled once, bounded to $[0.3, 3.0]\times$ the initial value. This adapts bidirectionally without manual scheduling.

Adaptive inner recovery. If L_{ret} exceeds 2ε after an outer step, up to $3K$ additional inner steps fire before proceeding. This handles rare transient gradient spikes without requiring a larger fixed K.

C Extended Results

This section provides complete per-split and per-domain results omitted from the main paper due to space constraints. Table 9 reports all TOFU 3B metrics across three forget splits with 5-fold standard deviations; Table 10 gives the corresponding LLM judge scores; and Table 11 reports KnowUndo results for both copyright and privacy domains. MUSE results are reported in full in the main paper (Table 2).

Key observations. On TOFU 3B, BLADE and PDU are the only methods that consistently achieve near-zero forget leakage (Prob $\leq$ 0.01) across all splits while maintaining utility. BLADE wins on the LLM judge metric (HM_J $\geq$ 0.962) because it achieves lower forget leakage scores without sacrificing response quality. On KnowUndo, BLADE achieves the highest HM on both domains (+0.021 over the next best on copyright, +0.059 on privacy) and the highest judge HM (+0.06 on copyright, +0.06 on privacy), demonstrating that the bilevel constrained approach generalizes beyond synthetic forget sets to real-world copyright and PII removal.

D Training Dynamics

Figure 5 shows BLADE dynamics on additional benchmarks beyond the MUSE Books example in the main paper. The three-phase pattern is consistent: (1) *warm-up*: forget loss drops rapidly while retain loss stays within ε; (2) *spike-and-ratchet*: transient retain violations trigger the dual ratchet, pushing λ up by discrete increments per Proposition 2; and (3) *smooth convergence*: λ stabilizes and both losses decrease monotonically. The final λ scales with task difficulty without any hyperparameter change, reflecting the harder optimization landscape when more knowledge must be

Table 9: TOFU (Llama-3.2-3B), 5 seeds. Best per column in **bold** (excl. Gold and collapsed models).

Split	Method	MU↑	Prob↓	ROUGE↓	HM↑
fgt01	Gold (retrain)	.663	.179	.409	.656
	GradAscent	.668$_{\pm .001}$	.564$_{\pm .009}$	.525$_{\pm .014}$	.509$_{\pm .008}$
	GradDiff	.659$_{\pm .003}$	.569$_{\pm .019}$	.578$_{\pm .027}$	.483$_{\pm .020}$
	NPO	.668$_{\pm .000}$	.553$_{\pm .009}$	.494$_{\pm .008}$	.525$_{\pm .005}$
	SimNPO	.656$_{\pm .002}$	.920$_{\pm .007}$	.830$_{\pm .021}$	.151$_{\pm .012}$
	RMU	.657$_{\pm .001}$	.861$_{\pm .002}$	.715$_{\pm .006}$	.246$_{\pm .003}$
	BLURNPO	.667$_{\pm .003}$	.822$_{\pm .015}$	.788$_{\pm .024}$	.253$_{\pm .021}$
	PDU	**.692**$_{\pm .001}$	.168$_{\pm .006}$	.283$_{\pm .007}$	.742$_{\pm .003}$
	BLADE	.654$_{\pm .003}$	**.000**$_{\pm .000}$	**.015**$_{\pm .011}$	**.846**$_{\pm .004}$
fgt05	Gold (retrain)	.659	.130	.388	.698
	GradAscent	.481$_{\pm .015}$	.140$_{\pm .015}$	.333$_{\pm .013}$	.632$_{\pm .008}$
	GradDiff	.566$_{\pm .005}$	.147$_{\pm .008}$	.395$_{\pm .010}$	.653$_{\pm .003}$
	NPO	.542$_{\pm .005}$	.219$_{\pm .004}$	.359$_{\pm .012}$	.640$_{\pm .003}$
	SimNPO	.657$_{\pm .001}$	.901$_{\pm .002}$	.818$_{\pm .005}$	.175$_{\pm .004}$
	RMU	.644$_{\pm .001}$	.602$_{\pm .004}$	.534$_{\pm .003}$	.483$_{\pm .002}$
	BLURNPO	.640$_{\pm .022}$	.683$_{\pm .082}$	.593$_{\pm .091}$	.412$_{\pm .075}$
	PDU	**.687**$_{\pm .001}$	.009$_{\pm .001}$	.085$_{\pm .009}$	**.843**$_{\pm .002}$
	BLADE	.655$_{\pm .002}$	**.007**$_{\pm .006}$	**.028**$_{\pm .016}$	.842$_{\pm .005}$
fgt10	Gold (retrain)	.661	.115	.382	.698
	GradAscent	.000$_{\pm .000}$	.000$_{\pm .000}$	.003$_{\pm .004}$	.000$_{\pm .000}$
	GradDiff	.532$_{\pm .015}$	.079$_{\pm .008}$	.361$_{\pm .024}$	.662$_{\pm .005}$
	NPO	.549$_{\pm .013}$	.237$_{\pm .005}$	.353$_{\pm .059}$	.640$_{\pm .015}$
	SimNPO	.652$_{\pm .001}$	.889$_{\pm .002}$	.808$_{\pm .004}$	.191$_{\pm .003}$
	RMU	.647$_{\pm .001}$	.390$_{\pm .010}$	.472$_{\pm .002}$	.591$_{\pm .004}$
	BLURNPO	.385$_{\pm .167}$	.245$_{\pm .163}$	.314$_{\pm .094}$	.502$_{\pm .099}$
	PDU	**.680**$_{\pm .009}$	**.000**$_{\pm .000}$	**.029**$_{\pm .008}$	**.857**$_{\pm .003}$
	BLADE	.647$_{\pm .004}$	.005$_{\pm .003}$	.038$_{\pm .014}$	.836$_{\pm .006}$

Table 10: LLM Judge on TOFU 3B (5 seeds, Claude Opus 4.7). Best $\mathrm{HM_J}$ in **bold**.

Split	Method	FL↓	RA↑	rRQ↑	$\mathrm{HM_J}$↑
fgt01	GradAscent	1.335$_{\pm .086}$	1.849$_{\pm .003}$	1.998$_{\pm .000}$	.587$_{\pm .045}$
	GradDiff	1.500$_{\pm .066}$	1.818$_{\pm .017}$	1.997$_{\pm .001}$	.490$_{\pm .041}$
	NPO	1.230$_{\pm .041}$	1.843$_{\pm .004}$	1.998$_{\pm .001}$	.640$_{\pm .019}$
	SimNPO	1.825$_{\pm .053}$	1.839$_{\pm .008}$	1.995$_{\pm .001}$	.220$_{\pm .057}$
	RMU	1.585$_{\pm .038}$	1.807$_{\pm .002}$	1.995$_{\pm .001}$	.432$_{\pm .027}$
	BLURNPO	1.730$_{\pm .122}$	1.853$_{\pm .003}$	1.998$_{\pm .001}$	.308$_{\pm .109}$
	PDU	0.615$_{\pm .042}$	1.722$_{\pm .007}$	1.968$_{\pm .004}$	.828$_{\pm .011}$
	BLADE	**0.010**$_{\pm .014}$	1.845$_{\pm .004}$	1.995$_{\pm .001}$	**.970**$_{\pm .002}$
fgt05	GradAscent	0.693$_{\pm .042}$	1.452$_{\pm .021}$	1.981$_{\pm .012}$	.766$_{\pm .009}$
	GradDiff	0.833$_{\pm .007}$	1.293$_{\pm .022}$	1.705$_{\pm .062}$	.677$_{\pm .009}$
	NPO	0.738$_{\pm .037}$	1.552$_{\pm .009}$	1.995$_{\pm .001}$	.774$_{\pm .009}$
	SimNPO	1.742$_{\pm .013}$	1.854$_{\pm .007}$	1.996$_{\pm .001}$	.305$_{\pm .012}$
	RMU	1.107$_{\pm .033}$	1.703$_{\pm .008}$	1.991$_{\pm .003}$	.679$_{\pm .014}$
	BLURNPO	1.397$_{\pm .153}$	1.712$_{\pm .054}$	1.997$_{\pm .001}$	.538$_{\pm .078}$
	PDU	0.067$_{\pm .013}$	1.749$_{\pm .009}$	1.978$_{\pm .003}$	.941$_{\pm .002}$
	BLADE	**0.023**$_{\pm .026}$	1.817$_{\pm .011}$	1.990$_{\pm .004}$	**.962**$_{\pm .004}$
fgt10	GradAscent	0.000$_{\pm .000}$	0.000$_{\pm .000}$	0.000$_{\pm .000}$	.000$_{\pm .000}$
	GradDiff	0.663$_{\pm .029}$	1.164$_{\pm .058}$	1.432$_{\pm .199}$	.648$_{\pm .037}$
	NPO	0.613$_{\pm .078}$	1.519$_{\pm .042}$	1.992$_{\pm .006}$	.796$_{\pm .011}$
	SimNPO	1.713$_{\pm .006}$	1.858$_{\pm .005}$	1.995$_{\pm .002}$	.331$_{\pm .006}$
	RMU	0.851$_{\pm .017}$	1.703$_{\pm .006}$	1.989$_{\pm .002}$	.765$_{\pm .005}$
	BLURNPO	0.726$_{\pm .354}$	1.046$_{\pm .508}$	1.508$_{\pm .601}$	.550$_{\pm .161}$
	PDU	0.014$_{\pm .005}$	1.726$_{\pm .029}$	1.952$_{\pm .038}$	.940$_{\pm .011}$
	BLADE	0.034$_{\pm .028}$	1.837$_{\pm .011}$	1.992$_{\pm .003}$	**.965**$_{\pm .004}$

Table 11: KnowUndo full results (5-fold). HM = hmean(1−fgt_R, ret_R, MMLU). Judge HM = hmean(1−FL/2, RA/2, rRQ/2).

Method	Copyright						LLM Judge			
	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑	MMLU↑	HM↑	FL↓	RA↑	rRQ↑	HM↑
FT (target)	.854	.244	.894	.336	.439	.456	1.50	1.80	1.13	.44
GradAscent	.002	.000	.002	.000	.250	.000	0.00	0.00	0.00	.00
GradDiff	.002	.000	.642	.199	.386	.348	0.00	0.81	0.53	.41
NPO	.678	.189	.763	.289	.445	.433	0.77	1.74	1.15	.66
SimNPO	.303	.048	.788	.317	.435	.461	0.05	1.73	1.12	.75
RMU	.435	.076	.763	.201	.388	.347	0.04	1.01	0.98	.60
PDU	.045	.011	.836	.285	.442	.442	0.01	1.59	0.93	.68
BLADE	.332	**.040**	**.856**	**.332**	**.449**	**.477**	**0.04**	**1.79**	**1.17**	**.78**

Method	Privacy						LLM Judge			
	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑	MMLU↑	HM↑	FL↓	RA↑	rRQ↑	HM↑
FT (target)	.940	.665	.949	.698	.445	.450	1.82	1.70	1.95	.23
GradAscent	.017	.004	.012	.006	.318	.017	0.00	0.00	0.00	.00
GradDiff	.049	.039	.708	.406	.428	.513	0.04	0.92	1.23	.62
NPO	.646	.322	.790	.426	.450	.496	0.74	1.12	1.99	.68
SimNPO	.516	.302	.889	.557	.438	.544	0.58	1.42	1.88	.77
RMU	.568	.072	.661	.079	.361	.182	0.07	0.07	0.21	.08
PDU	.055	.022	.813	.452	.444	.546	0.04	1.09	1.50	.72
BLADE	.306	.128	**.937**	**.608**	**.462**	**.605**	0.27	**1.46**	**1.87**	**.83**

erased.

Figure 6 shows PDU's contrasting behavior. The unbounded logit-margin loss and symmetric dual updates produce recurring retain oscillations that never fully converge, with amplitude growing on larger forget sets. This instability explains PDU's collapse at 4× scale in the scalability experiment (Section 5.5).

Figure 7 shows the inner loop ablation across $K \in \{0, 3, 6\}$ and 8 ε-multiplier settings. K=3 consistently outperforms K=0, confirming that the inner loop is necessary for retain protection. Doubling to K=6 yields no further gain while being substantially slower per step, demonstrating clear diminishing returns and confirming K=3 as the efficiency sweet spot.

E Qualitative Examples

We present representative generations (seed=42) illustrating the forget–retain tradeoff on each benchmark. Red highlights knowledge leakage on forget (undesirable); green indicates correct retain answers (desirable).

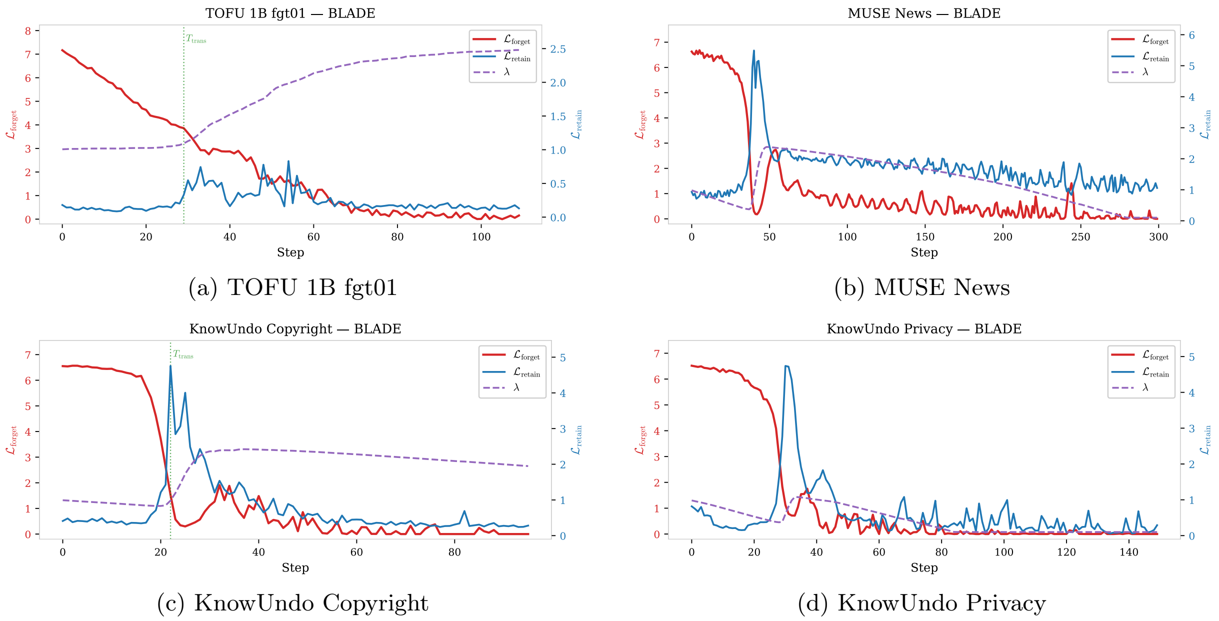
On non-linguistic outputs. BLADE's clamped entropy objective targets high distributional uncertainty on forget tokens, which can produce non-linguistic outputs rather than coherent refusals. We argue this is preferable to the alternative exhibited by baselines: fluent, confident hallucinations that partially or fully reproduce the target content. Recent work demonstrates that LLMs readily amplify plausible-sounding misinformation—even fabricated medical conditions are absorbed and confidently presented as fact (Stokel-Walker, 2026)—creating a dangerous feedback loop when downstream users treat fluent outputs as authoritative. In the context of unlearning (copyrighted material, private data, hazardous knowledge), a non-linguistic response *signals* that the model lacks the requested knowledge, whereas a fluent hallucination risks laundering prohibited content in paraphrased form that evades detection. Crucially, BLADE's retain performance confirms that this behavior is selective: the model remains fully coherent on non-forget inputs (Tables 13, 15, 17), demonstrating that the non-linguistic outputs reflect targeted forgetting rather than model corruption.

F Use of AI Assistants

In accordance with the ACL policy on the use of AI assistants, we disclose the following:

Tool. Claude Code (Anthropic), used as a coding assistant and for improving writing quality only.

Scope of use. Claude Code was used for: (1) implementation of training loops, eval-



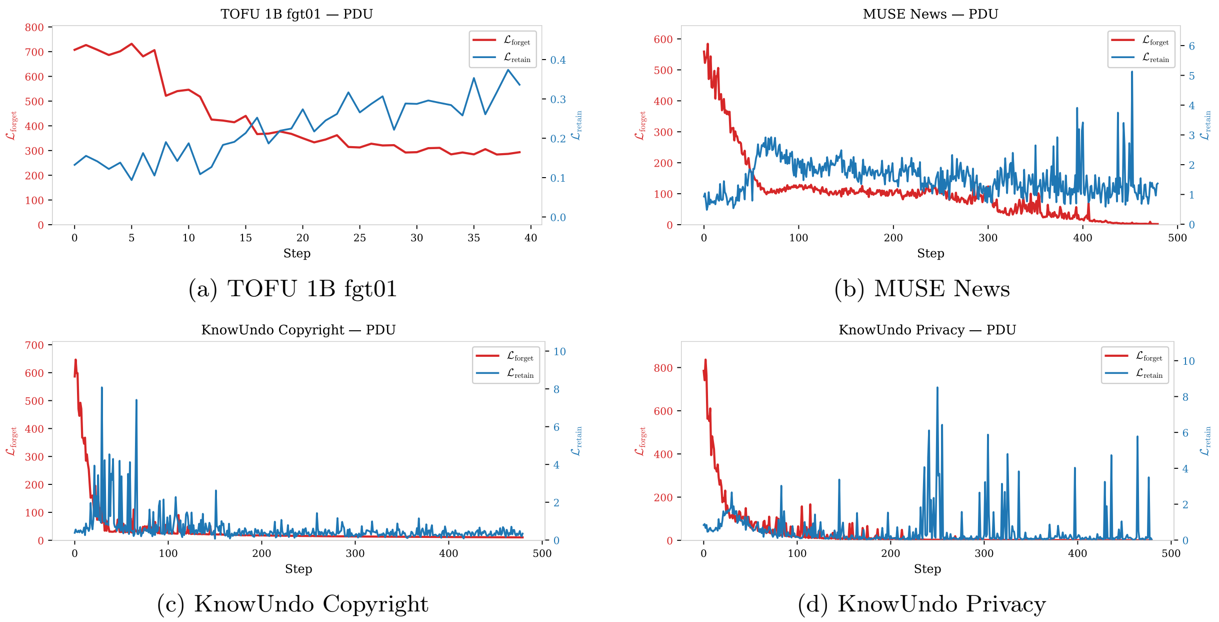
(a) TOFU 1B fgt01

(b) MUSE News

(c) KnowUndo Copyright

(d) KnowUndo Privacy

Figure 5: BLADE training dynamics across all benchmarks. The three-phase pattern is consistent; λ adapts to task difficulty without hyperparameter change. MUSE Books is shown in Figure 4.



(a) TOFU 1B fgt01

(b) MUSE News

(c) KnowUndo Copyright

(d) KnowUndo Privacy

Figure 6: PDU training dynamics across all benchmarks. Unbounded logit-margin loss and symmetric dual updates produce recurring retain oscillations across all settings.

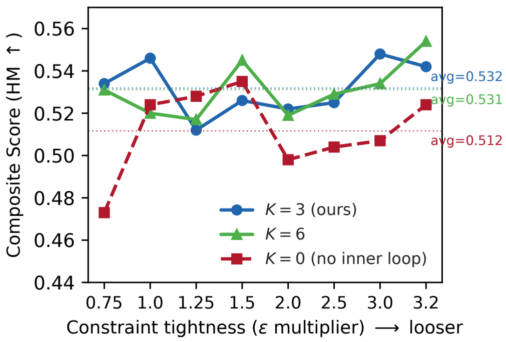


Figure 7: Inner loop ablation on MUSE News across 8 ε-multiplier settings. K=3 consistently outperforms K=0; K=6 provides no additional gain.

Table 12: TOFU **forget**: most baselines reproduce the memorized answer; BLADE produces non-linguistic tokens.

Q: What gender is author Basil Mahfouz Al-Kuwaiti?
Gold: *Basil Mahfouz Al-Kuwaiti is male.*

GradAscent	Basil Mahfouz Al-Kuwaiti is male.
GradDiff	Basil Mahfouz Al-Kuwaiti is male.
NPO	Basil Mahfouz Al-Kuwaiti is male.
SimNPO	Basil Mahfouz Al-Kuwaiti is male.
RMU	Basil Mahfouz Al-Kuwaiti is male.
BLURNPO	Basil Mahfouz Al-Kuwaiti is male.
PDU	Basil is a male, from...Bahrain
BLADE	yourselvesilden... *(non-linguistic)*

Table 13: TOFU **retain**: BLADE perfectly retains knowledge that PDU degrades. All other methods also retain correctly on this example.

Q: What inspired Anara Yusifova to become an author?
Gold: *...inspired by her love for storytelling and her desire to share...*

GradAscent	...inspired by her love for storytelling...
NPO	...inspired by her love for storytelling...
SimNPO	...inspired by her love for storytelling...
RMU	...inspired by her love for storytelling...
BLURNPO	...inspired by her love for storytelling...
PDU	Inspired by her mother's investigative journalism...
BLADE	...inspired by her love for storytelling...

uation scripts, and figure-generation code;

Table 14: MUSE Books **forget**: NPO and PDU fully reproduce the target; BLADE outputs a non-memorized token.

Q: Who was arrested for the murders, according to the Riddles' cook?
Gold: *Frank Bryce*

GradAscent	*(collapsed)*
GradDiff	*(collapsed)*
NPO	Frank Bryce
SimNPO	'Frank'
RMU	Dobby
PDU	Frank Bryce
BLADE	197

Table 15: MUSE Books **retain**: only BLADE answers correctly; all other methods (including PDU, NPO, RMU) fail.

Q: Who did Ginny recommend to accompany Harry in Cho's place?
Gold: *Luna Lovegood*

GradAscent	*(collapsed)*
GradDiff	*(collapsed)*
NPO	Hermione Granger
SimNPO	Hermione
RMU	Michael Corner
PDU	Jacob's sibling
BLADE	Luna Lovegood

Table 16: KnowUndo copyright **forget**: only NPO reproduces content matching the fine-tuned target; BLADE outputs non-linguistic tokens.

Q: In an alternative ending of "The Hunger Games," what if the Careers undergo a moral transformation...?
Gold: *...the Careers undergo a moral transformation and ally with...*

GradAscent	*(collapsed)*
GradDiff	*(collapsed)*
NPO	...the Careers undergo a transformation...
SimNPO	*(collapsed)*
RMU	rock music.
PDU	*(collapsed)*
BLADE	*(non-linguistic tokens)*

(2) stylometric improvements to text written by the authors, without adding or removing any scientific content, claims, or conclusions. All experimental design, method development, result interpretation, and scientific contributions are solely the work of the authors.

Verification. All code outputs were reviewed and validated by the authors. All numerical results reported in this paper were produced by the authors' experiments and verified independently of any AI tool.

Table 17: KnowUndo copyright **retain**: BLADE produces the highest-quality answer; GradAscent collapses and RMU/PDU degrade.

Q: I enjoyed "Harry Potter." Can you suggest similar series? Gold: *If you enjoy the magical and adventurous elements of "Harry Potter"...*	
GradAscent	*(Cyrillic garbage — collapsed)*
GradDiff	If you enjoy the magical and adventurous...
NPO	If you enjoy the magical and adventurous...
SimNPO	If you enjoyed the magical and adventurous...
RMU	If you're looking for books similar to...
PDU	If you're a fan of "Harry Potter" and...
BLADE	If you enjoy the magical and adventurous...